

LAB2.md

Script 1: print_numbers.sh

Purpose:

Prints numbers from 1 to 5.

Code:

```
#!/bin/bash
for i in {1..5}
do
    echo $i
done
```

Line-by-line Explanation:

1. `#!/bin/bash` — Tells the system to use the Bash shell to run this script.
2. `for i in {1..5}` — Loops through numbers 1 to 5.
3. `do` — Starts the loop body.
4. `echo $i` — Prints the current number.
5. `done` — Ends the loop.

Example Run:

```
$ ./print_numbers.sh
1
2
3
4
5
```

```
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % nano backup.sh  
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % chmod 777 backup.sh  
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % ./backup.sh  
1  
2  
3  
4  
5  
vaibhavkumar@VAIBHAVs-MacBook-Air ~ % █
```

```
}  
}  
}
```

Script 2: array_loop.sh

Purpose:

Prints each element of an array.

Code:

```
#!/bin/bash  
arr=(apple banana cherry)  
for fruit in "${arr[@]}"  
do  
    echo $fruit  
done
```

Line-by-line Explanation:

1. `#!/bin/bash` — Uses Bash shell.
2. `arr=(apple banana cherry)` — Declares an array with three fruits.
3. `for fruit in "${arr[@]}"` — Loops through each element in the array.
4. `do` — Starts the loop body.
5. `echo $fruit` — Prints the current fruit.
6. `done` — Ends the loop.

Example Run:

```
$ ./array_loop.sh  
apple  
banana  
cherry
```

```
(vaibhavkumar@VAIBHAVs-MacBook-Air scripts % nano scripts.sh
vaibhavkumar@VAIBHAVs-MacBook-Air scripts % ./scripts.sh
apple
banana
cherry
vaibhavkumar@VAIBHAVs-MacBook-Air scripts %
```

Extra Questions

1. What is the purpose of `#!/bin/bash` at the top of a script?

It tells the system to use the Bash shell to interpret and run the script.

2. How do you make a script executable?

Run:

```
chmod +x scriptname.sh
```

Replace `scriptname.sh` with your script's filename.